Bridging Mathematics

Fall 2025

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S17-08-23, Department of Mathematics
Office Hours by Appointment

In this course, we take a deep dive into three main areas of mathematics: we decode the mysteries of change in *calculus*, we unlock the structures of space in *linear algebra*, and we make sense of randomness in *probability*. By the end of this course, you will be equipped with the core skills and foundational knowledge needed to confidently tackle more advanced courses in the sciences and engineering.

The course runs for all thirteen weeks of the semester, and all lectures and tutorials will be held in person.

Four 'booster' lectures will be conducted in weeks 1 and 2 to launch the course, during which we will review fundamental concepts in calculus. I will use this as an opportunity to gauge the mathematical ability of the class so as to determine the pace and difficulty of the remainder of the course. From week 3 to week 13, lectures will only be held every Wednesday; Mondays will instead be dedicated to tutorials.

In-class lecture time is devoted to a formal introduction of each week's topics, where we begin to grapple with new ideas and gain exposure to a variety of problems. These ideas are more deeply examined during the tutorial, where we will hone your problem-solving skills through a close study of problems involving the topics discussed during the lecture.

You are expected to learn each week's material in a timely manner. Everyone is very much encouraged to interact with the instructor and with each other: ask questions and clarify concepts as and when you encounter them. You are also expected to come prepared to each week's tutorial: if not with full solutions to the problems, then with questions about the material and attempts at the solutions. Your questions will not only help you identify and reflect on the problems you find difficult, but also drive our discussion in class.

Your progress will be assessed via two main checkpoints.

- The **First Checkpoint** will (tentatively) be held on the 29th of September (Monday of Week 7). This checkpoint will cover the material discussed during the first half of the course, including single-variable calculus, calculus on vector functions, and basic multivariable calculus.

- The **Second Checkpoint** will (tentatively) be held on the 24th of November (Monday of Examination Week 1). This checkpoint is comprehensive in scope and will cover everything discussed throughout the course, but it will mainly test you on ideas and techniques from linear algebra and probability.

Prior to each of these checkpoints, a **Mock Test** will be provided to help you prepare the exam and gain a sense of how they will be structured.

Class Schedule

Week 1: A Review of Calculus, Part I (11 August, 13 August)

Topics Functions of a single variable. Limits. Differentiation and its applications. Integration and its applications.

To-Do Complete the *Calculus Review*, Part 1.

Week 2: A Review of Calculus, Part II (18 August, 20 August)

Topics Techniques of integration. First-order differential equations. Infinite sequences and series.

To-Do Complete the *Calculus Review*, Part 2.

Week 3: The Geometry of Space (25 August, 27 August)

Topics Vectors in three dimensions. The dot product and the cross product. Equations of lines and planes.

To-Do Complete **Tutorial 1** on Infinite Sequences and Series.

Week 4: Vector Functions (1 September, 3 September)

Topics Parametrized curves and surfaces. Calculus on vector functions. Arc length and curvature.

To-Do Complete **Tutorial 2** on the Geometry of Space.

Week 5: An Introduction to Multivariable Calculus (8 September)

Topics Functions of several variables. Partial derivatives. Multiple integrals.

To-Do There is no tutorial this week.

Week 6: Calculus Retrospective (15 September, 17 September)

Topics A review of multivariable calculus.

To-Do Complete **Tutorial 3** on Multivariable Calculus.

Recess Week

To-Do Attempt **Mock Test 1**. Prepare for the **First Checkpoint**.

Week 7: An Introduction to Linear Algebra (29 September, 1 October)

Topics Systems of linear equations. Gaussian elimination. Matrices and linear transformations.

To-Do Complete the **First Checkpoint**.

Week 8: Subspaces of $\mathbb{R}^n$ (6 October, 8 October)

Topics Vectors in Euclidean space. The image and kernel of a linear transformation. Bases, coordinates, and dimension.

To-Do Complete **Tutorial 4** on Linear Systems.

Week 9: Orthogonality (13 October, 15 October)

Topics Orthogonal projections. Least squares and data fitting. A review of linear algebra.

To-Do Complete **Tutorial 5** on Vectors in Euclidean Space.

Week 10: An Introduction to Probability (22 October)

Topics Basic combinatorics. The axioms of probability. Conditional probability.

To-Do There is no tutorial this week. In your own time, attempt the *Linear Algebra Review*.

Week 11: Random Variables (27 October, 29 October)

Topics Expected value and variance. Discrete random variables. Continuous random variables. Probability models.

To-Do Complete **Tutorial 6** on the Fundamentals of Probability.

Week 12: Sampling Distributions (3 November, 5 November)

Topics Sampling distribution models. The Central Limit Theorem. Confidence intervals. Hypothesis testing.

To-Do Complete **Tutorial 7** on Random Variables.

Week 13: A Look Back on the Course (10 November, 12 November)

Topics A review of probability.

To-Do Complete **Tutorial 8** on Sampling Distributions.

Reading Week

To-Do Attempt **Mock Test 2**. Prepare for the **Second Checkpoint**.

Examination Week

To-Do Complete the **Second Checkpoint**.